# Escape the Geese – CRC cards

Cohort 2 Team 3 – ‘Kostas Barmpis and Dimitris Kolovos’

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## Screens

| Main (extends LibGDX) | |
|---|---|
| Responsibility | Collaborator |
| Start the application and load assets once | Settings |
| Show the menu first | MenuScreen |
| Switch to the tutorial, game or end screen on request | TutorialScreen, GameScreen, EndScreen |
| Hold the shared SpriteBatch and Settings for every screen | Settings |

| MenuScreen | |
|---|---|
| Responsibility | Collaborator |
| Show the Title, Play, Tutorial and Quit | |
| Start a new game | Main, GameScreen |
| Open a tutorial | Main, TutorialScreen |

| TutorialScreen | |
|---|---|
| Responsibility | Collaborator |
| Show controls, item effects, the exhaustion meter and the goal | |
| Return to the menu or start the game | Main, MenuScreen |

| GameScreen | |
|---|---|
| Responsibility | Collaborator |
| Create one session: map, player, waves, events, map items | GameMap, Player, WaveManager, Event, Item |
| Run the frame loop | Player, WaveManager, Event, HUD |
| Start paused; pause and resume; show the pause menu | GameTimer, Settings |
| Pass key presses to the player | Player |
| Check each event's trigger zone against the player's position. | Event, Player |
| Check win (all waves cleared) and loss (timer or meter) | WaveManager, GameTimer, ExhaustionMeter |
| Hand the result and score to the end screen | Main, EndScreen, Score |

| EndScreen | |
|---|---|
| Responsibility | Collaborator |
| Show win or loss and the reason (cleared, time up, exhausted) | |
| Show final score, items used and events triggered | Score |
| Restart or return to the menu | Main |

## Gameplay

| Player | |
|---|---|
| Responsibility | Collaborator |
| Move with WASD or arrows | |
| Stay inside the map and off wall | GameMap |
| Stop when a goose blocks the way | Goose |
| Pick up items walked over | Item, Inventory |
| Use or throw the selected item | Inventory, Item |
| Own the exhaustion meter | ExhaustionMeter |

| ExhaustionMeter | |
|---|---|
| Responsibility | Collaborator |
| Fill while the geese are near or blocking; recover slowly otherwise. | Goose, Player |
| Report when full (loss) | GameScreen |
| Expose its level for drawing | HUD |

| Inventory | |
|---|---|
| Responsibility | Collaborator |
| Holds items; starts with 5 | Item |
| Track the selected slot | Player |
| Remove an item once used | Item |
| Expose icons and counts for drawing | HUD |

| Item | |
|---|---|
| Responsibility | Collaborator |
| Know name, icon, range and strength | |
| Apply its effect to geese within range where it lands | Goose |
| Breadcrumbs: disperse all geese in an area | Goose |
| Pebble: distract a goose if thrown near; enrage it if thrown at it | Goose |
| Report a dispersal for points | Score |

| Goose | |
|---|---|
| Responsibility | Collaborator |
| Know position, speed and state | |
| Chase the player around walls | Player, GameMap |
| Block the player and fill the meter when close | Player, ExhaustionMeter |
| React to items: distracted, enraged or dispersed | Item |
| React to events | Event |
| Leave the map when dispersed and tell its wave | WaveManager |

| WaveManager | |
|---|---|
| Responsibility | Collaborator |
| Spawn each wave of geese at the map edges | Goose, GameMap |
| Run the short countdown between waves | HUD |
| Track geese left; report a cleared wave | Score, GameScreen |
| Report when every wave is cleared (win) | GameScreen |
| Change the next wave's size when event says so | Event |

| Event (abstract; subclasses ScoreBoost, GooseRush) | |
|---|---|
| Responsibility | Collaborator |
| Know its trigger zone, duration and whether it is used | |
| Trigger once when the player enters the zone | GameScreen, Player |
| Play its sound and show its visual cue | Settings |
| Apply its effect: geese, player, waves or score | Goose, Player, WaveManager, Score |
| Add or remove items on the map | Item |

| Score | |
|---|---|
| Responsibility | Collaborator |
| Add points for each dispersal, more early in a wave | WaveManager, Item |
| Apply and expire multipliers | Event |
| Count items used and events triggered | Player, Event |
| Expose totals for HUD and the end screen | HUD, EndScreen |

## Support

| GameMap | |
|---|---|
| Responsibility | Collaborator |
| Load and draw the campus map with landmarks | |
| Answer whether a position can be walked on | Player, Goose |
| Give spawn points and interactable | WaveManager, GameScreen |

| GameTimer | |
|---|---|
| Responsibility | Collaborator |
| Count down 5 minutes; stop while paused | GameScreen |
| Report when time is up (loss) | GameScreen |
| Expose time left for drawing | HUD |

| HUD | |
|---|---|
| Responsibility | Collaborator |
| Draw time, score, items, meter, wave and counters | GameTimer, Score, Inventory, ExhaustionMeter, WaveManager |
| Stay readable windowed and full screen | Setting |

| Setting | |
|---|---|
| Responsibility | Collaborator |
| Hold volume and full screen state | |
| Apply fullscreen when toggled | Main |
| Give the current volume to sounds | Event |